

INVITED TALK / ABSTRACT

On a Fully Nonlinear Conformal Flow

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Abstract

The k -Yamabe flow is a fully nonlinear conformal flow that arises naturally in conformal geometry. In this talk, we study radially symmetric gradient k -Yamabe solitons in the locally conformally flat setting, focusing on their existence, classification, and asymptotic behavior. These special solutions play a central role in the analysis of the flow and extend the classification results of P. Daskalopoulos and N. Sesum to the case $n > 2k$. We will also discuss ongoing work on the existence of such solutions in the negative cone, with particular emphasis on understanding their non-admissibility. Joint work with Mariel Sáez.

Keywords: fully nonlinear conformal flow; k -Yamabe flow; conformal geometry; gradient k -Yamabe solitons; locally conformally flat setting; existence; classification; asymptotic behavior; negative cone